

$Q = Q_1 \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5$  true.

• When  $n=8$  there are 92 such placements, while for  $n=16$  this number grows to 14,772,512.

### Exercise 1.3

1. Use truth tables to verify these equivalences.

a)  $p \wedge T \equiv p$

| p | T | $p \wedge T$ |
|---|---|--------------|
| T | T | T            |
| F | T | F            |

b)  $p \vee F \equiv p$

| p | F | $p \vee F$ |
|---|---|------------|
| T | F | T          |
| F | F | F          |

c)  $p \wedge F \equiv F$

| p | F | $p \wedge F$ |
|---|---|--------------|
| T | F | F            |
| F | F | F            |

d)  $p \vee T \equiv T$

| p | T | $p \vee T$ |
|---|---|------------|
| T | T | T          |
| F | T | T          |

e)  $p \vee p \equiv p$

| p | $p \vee p$ |
|---|------------|
| T | T          |
| F | F          |

f)  $p \wedge p \equiv p$

| p | $p \wedge p$ |
|---|--------------|
| T | T            |
| F | F            |

2. Show that  $\neg(\neg p)$  and  $p$  are logically equivalent

| p | $\neg p$ | $\neg(\neg p)$ |
|---|----------|----------------|
| T | F        | T              |
| F | T        | F              |

3. Use truth tables to verify the commutative laws

a)  $p \vee q \equiv q \vee p$

| p | q | $p \vee q$ | $q \vee p$ |
|---|---|------------|------------|
| T | T | T          | T          |
| T | F | T          | T          |
| F | T | T          | T          |
| F | F | F          | F          |

b)  $p \wedge q \equiv q \wedge p$

| p | q | $p \wedge q$ | $q \wedge p$ |
|---|---|--------------|--------------|
| T | T | T            | T            |
| T | F | F            | F            |
| F | T | F            | F            |
| F | F | F            | F            |

4. Use truth tables to verify the associative laws

a)  $(p \vee q) \vee r \equiv p \vee (q \vee r)$

b)  $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$